Dmitry Messerman

Senior VLSI CAD Engineer
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Professional Summary

- **Power and signal integrity tools expert** with over 10 years of experience in development of power and signal integrity solutions for advanced silicon package and printed circuit board.
- **EDA CAD tool expert** with over 10 years of experience in the development, support, and deployment of backend CAD and Silicon Debug tools for high-speed and analog custom design.
- **Developed and delivered career development mentoring classes** to experienced engineers.
- **Renowned for multi-disciplinary expertise, exceptional debugging and problem-solving skills.**

Technical Skills

Programming Languages: Python, C/C++, Perl, Tcl, Skill, MATLAB
Multiple Cadence, Synopsys and Siemens tools for Power and Signal Integrity, VLSI design and verification

Work Experience

2025 : Consultant, Anadeto Technology Ltd, Cyprus

- **Development of AI-based layout design tool**
  - Development of demo version and presentation to potential investor

2018 – 2025: Principal Engineer, Intel Corporation

- **Defining, co-developing and deploying multi-technology simulation** for Analog Custom Design
  - Solution integrates multiple blocks employing technologies supplied by Intel Foundry and leading 3rd party foundry
- **Defining a comprehensive QA platform** for CAD Analog design solutions
  - Platform supports more than 30 in-house and vendor capabilities integrated in RTL2GDS flow for Analog Design
- **Leading customization, QA effort and Customer support** of Design Engineers and Layout Designers in Analog design domain
  - Deployment of the above solution led to a drastic reduction in SW defects thus contributing to significant increase in design engineers and layout designers productivity
- **Developing and delivering career development mentoring classes** to experienced engineers

1998 - 2018: CAD SW Engineering Team Lead, Intel Corporation (*Principal Engineer from 2006*)

- **Leading development of in-house Power and Signal Integrity solution** for silicon package and printed circuit board.
  - Developing Fast Signal to Noise Ratio (SNR) estimation algorithm for PCB and silicon package interconnect
- **Leading development of post-silicon debug tools**:
  - **Platform HW/SW co-simulation** for BIOS MRC training validation
  - **Structure-Based Functional Tests (SBFT) coverage improvement** based on genetic algorithm

- **Leading definition, architecture and implementation of in-house Parasitics Extraction and Reliability verification tools**
  - Parasitics extraction tool has been **used in production for ~10 years**
  - Reliability verification tool has been **used in production for ~20 years**

- **Developing modeling algorithms in multiple areas:**
  - **Accurate and fast resistance extraction from VLSI layout** - computational geometry algorithm, **patent granted**
  - **Surrogate modeling of silicon heat dissipation in tester environment** based on diffusion equations solver

Education

Ph.D. in Electrical Engineering, St. Petersburg Polytechnic University, Russian Federation.
Developed 2-D Maxwell full-wave solver for high voltage transmission line above two-layer ground

M.Sc. in Electrophysics, St. Petersburg Polytechnic University, Russian Federation.

Publications and patents

1. Yan J., Aydiner A., **Messerman D**., Zhu J., Camacho Mora A., Liao J., Ma M., Jiao D. Efficient Equalization Optimization Algorithm for Signaling Analysis. Nonlinear System 2019 IEEE MTT-S International Conference on Numerical Electromagnetic and Multiphysics Modeling and Optimization (NEMO).
2. Reisfeld G., **Messerman D.**, Bone N., Lazar A. VARAN: variability analysis for memory cell robustness. Proc. SPIE 6925, 69250L, 2008.
3. **Messerman D.**, Seltser M. Reorganizing rectangular layout structures for improved extraction. - United States Patent 7389001 B2, 2008.
4. Chakravarty S., Ben-Noon Y., Chiprout E., Mazumder M., **Messerman D.** Inductance Modeling. - United States Patent 7325208 B2, 2008.

Languages

English – full professional proficiency, **Hebrew** - fluent, **Russian** – native.